An Agent Lab for Scientific Software Optimization

Deforming Bubble: FP64 to ready-to-evolve

Agent Lab

2026-10-20

The ask

- App: Deforming Bubble, level-set mini-app ported from Flash-X (WENO5 advection, redistance, MPI).
- Scientist: "I have added my code to the app, and ready for optimization."
- Agent: not yet — optimization needs a low-precision seed, so propose *raptorize*.
- App checked against 13-point contract: all pass, natively and in Docker.
- Plan ready 14 minutes later, every decision cited.

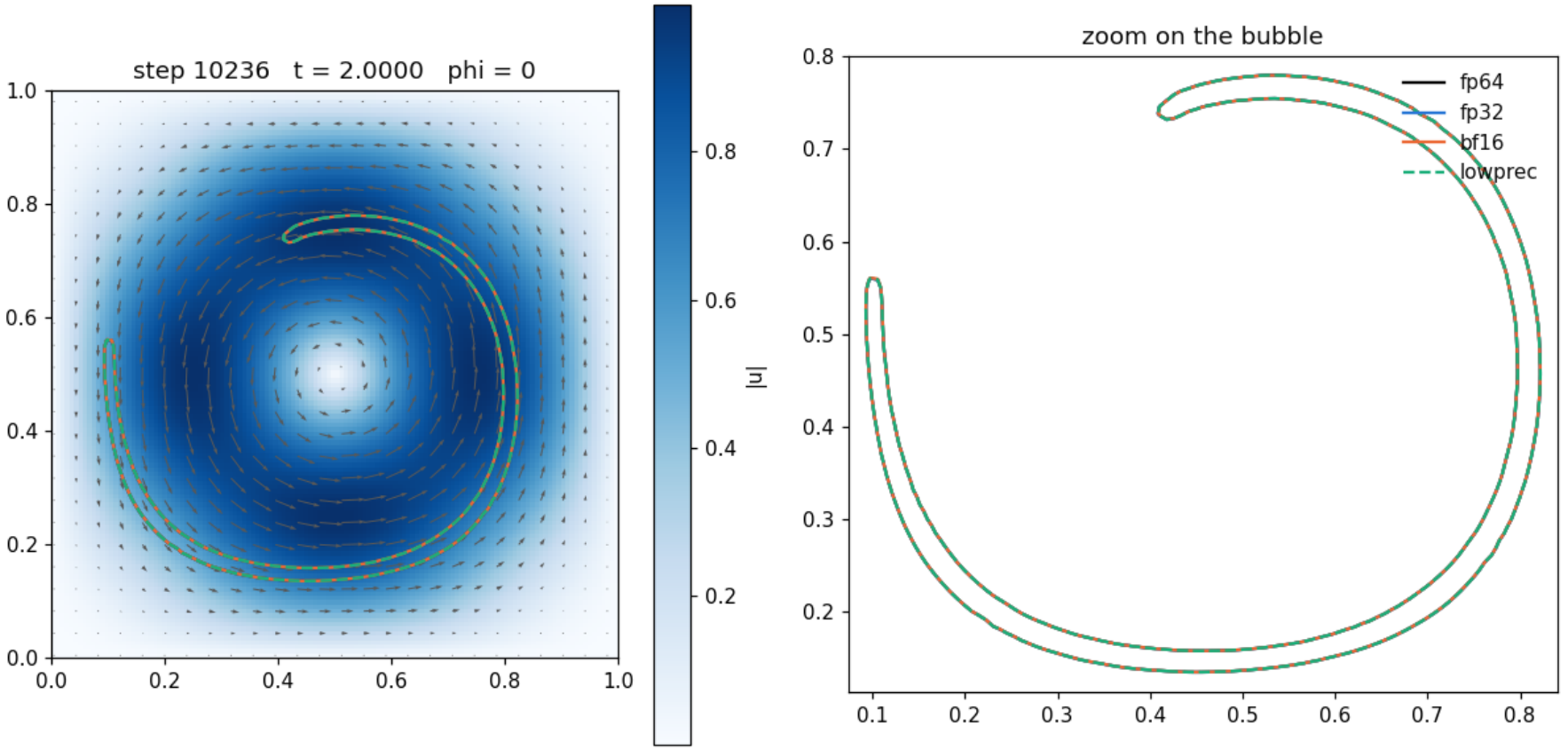
Lab layout

- `app/` swappable code (today: the bubble).
- `dev/workflows/` general contracts: protocol, raptorize, optimize, verify.
- `experiments/2026-10-20/`: question, runs, patches, results.
- Stages: raptorize $\rightarrow$ optimize $\rightarrow$ verify.

How it went

- 17 messages, 282 scientist words; 18 min scientist time.
- Profile: advection 52% runtime, redistance 24%.
- Baseline re-run in container so “unchanged” compares one compiler.
- Full run to deformation; MPI core-pinning fixed with one flag.
- 2 MPI ranks: field bit-identical to 1 rank in all 11 frames.

Results at $t = 2$



Error summary

mode	RMS error at $t = 2$	note
fp32 (emulated)	6.3×10^{-7}	30 min emulation
float seed (native)	6.3×10^{-7}	same, bit for bit, 17 s
bf16 (emulated)	2.6×10^{-4}	400x worse, invisible

- Float seed matches emulated fp32 bit for bit.
- Error lives away from the bubble edge.
- Hint: score near the interface in optimize.

Optimize so far

- Plan, evaluator, and job written.
- Seed score: 9.2722 (twice, identical).
- Red-team: cheats score null or below seed.
- Launch blocked on Google Cloud credentials.

Cost

class	tokens	$/M	$
fresh input	208	4.00	0.00
cache writes	624,107	5.00	3.12
cache reads	13,294,158	0.20	2.66
output	90,268	20.00	1.81
total			7.59

- 95% of reads from cache.
- Cache writes are the biggest line; long pauses re-cache.

Human vs. agent time

- By hand (expert, new to lab): ≈10 h.
- With agent: 18 min scientist attention.
- ≈30x reduction for $7.59.
- Agent worked 49 of 95 minutes.

What we'd change

- Keep open gates visible.
- One-hour cache for half-hour runs.
- Score where the physics is (interface error).
- App should create output folder parents.

Current state

- app/ restored from raptorize.patch; health checks pass.
- Optimize calibrated (9.2722) and red-teamed.
- Launch blocked on GCP credentials and YAML placeholders.